

Regularized e-processes¹

Anytime valid inference with knowledge-based efficiency gains

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- Theoretical justification for most statistical methods assumes sample sizes are (or can safely be treated as) fixed
- In practice:
 - the sample sizes might not be fixed
 - but the same fixed-sample-size methods are used anyway
- Lack of validity $\rightsquigarrow$ replication crisis, etc
- Methods that are valid for data-driven sample sizes?
- Broadly referred to as *anytime-validity*³
- Can be achieved with *e-processes* (or *e-values*)

³e.g., Ramdas, Grünwald, Vovk & Shafer, *Stat Sci* 2023

- Goal: choose most efficient (“largest”) e-process
- There are limits to how efficient/large an e-process can be⁴
- We have an obligation to push these limits, but how?
- Suppose You have prior knowledge... *regularization!*
- It’s easy to “regularize” an e-process, but such tweaks will surely jeopardize the anytime-validity property

Key question.

How to use prior knowledge in a principled way to boost an e-process without breaking its anytime-validity?

⁴Grünwald et al (*JRSS-B* 2024), GRO = growth rate optimal

- Background:
 - e-processes
 - prior knowledge $\rightsquigarrow$ imprecise probability
- Knowledge-based regularized e-processes:
 - construction
 - generalized Ville's inequality, etc
 - empirical illustration
- Clinical trial application
- From e-processes to full-blown uncertainty quantification
 - possibility-theoretic formulation
 - reliable decision-making
- Application, cont.

- Data $Z^n = (Z_1, \dots, Z_n)$ in $\mathbb{Z}^n$, $n \geq 1$, iid
- Model $\{P_\omega : \omega \in \mathbb{O}\}$
 - a genuine statistical model w/ parameter space $\mathbb{O}$
 - or $\mathbb{O}$ is just an index for all prob's on $\mathbb{Z}$
- Ω denotes the uncertain/unknown “true” ω
- Quantity of interest: $\Theta = f(\Omega)$, taking values in $\mathbb{T}$
- Examples:
 - $\Theta = f(\Omega) := \text{mean of } P_\Omega$
 - $\Theta = f(\Omega) := \arg \min_t P_\Omega L_t$, risk minimizer
- Throughout: $f : \mathbb{O} \rightarrow \mathbb{T}$ is continuous

- $\epsilon_\omega(Z^n)$ is an *e-process* for ω if
 - $\epsilon_\omega(\square) \leq 1$
 - $\epsilon_\omega(Z^n) \geq 0$ for all (n, Z^n)
 - and $E_\omega\{\epsilon_\omega(Z^n) \mid Z^{n-1}\} \leq \epsilon_\omega(Z^{n-1})$
- Then, for all stopping times N :
 - $\sup_\omega E_\omega\{\epsilon_\omega(Z^N)\} \leq 1$
 - $\sup_\omega P_\omega\{\epsilon_\omega(Z^N) \geq \alpha^{-1}\} \leq \alpha \leftarrow \text{Ville's inequality}$
- E-process for $\theta \dots \epsilon_\theta(z^n) = \inf_{\omega: f(\omega)=\theta} \epsilon_\omega(z^n)$
- Anytime-valid confidence sets, say:
 - define $C_\alpha(z^n) = \{\theta : \epsilon_\theta(z^n) < \alpha\}$
 - then coverage bounds hold for all stopping times N ,

$$\sup_\omega P_\omega\{C_\alpha(Z^N) \not\ni f(\omega)\} \leq \alpha$$

- Simplest example of an e-process is a Bayes factor:

$$\epsilon_{\omega}(Z^n) = \frac{\int p_{\omega}(Z^n) g(w) dw}{p_{\omega}(Z^n)}, \quad \omega \in \mathbb{O}$$

- All e-processes have a “likelihood ratio” form
- Other constructions:
 - universal inference (Wasserman, Ramdas, etc)
 - reverse information projection (Grünwald, etc)
- Shameless plug:
 - nonparametric e-process using recursive mixture model fits⁵
 - e-processes when no likelihood function is available⁶

⁵Dixit and M., [arXiv:2309.13441](#)

⁶Dey, M., and Williams, [arXiv:2402.00202](#)

- Prior knowledge about Θ ...
 - doesn't identify a prior distribution
 - it helps You rule out some of the infinitely-many priors
- What's left is a collection, $\mathcal{Q}$, of “plausible priors”
- Prior knowledge $\rightsquigarrow$ an *imprecise probability dist'n* for Θ
- Summarize this imprecise prior probability via

$$\underline{Q}(H) = \inf_{Q \in \mathcal{Q}} Q(H) \quad \text{and} \quad \overline{Q}(H) = \sup_{Q \in \mathcal{Q}} Q(H), \quad H \subseteq \mathbb{T}$$

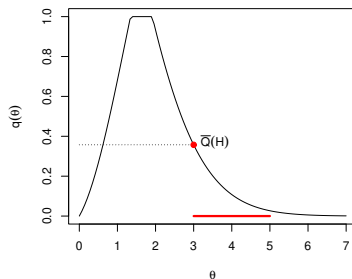
- Well-developed, mathematically rigorous theory of lower & upper probability/expectation: *Choquet integration*

Background, cont.

- My go-to: *possibility theory*
- Contour $q : \mathbb{T} \rightarrow [0, 1]$
- Encodes how *surprised* You'd be if " $\Theta = \theta$ " were true
 - large $q(\theta)$ if not surprised
 - small $q(\theta)$ if surprised
 - intermediate $q(\theta)$...
- Upper prob via optimization

$$\overline{Q}(H) = \sup_{\theta \in H} q(\theta)$$

- $\underline{Q}(H)$ is strictly less than $\overline{Q}(H)$



- Inputs:
 - data, model, etc
 - favorite e-process $\{\epsilon_\theta(\cdot) : \theta \in \mathbb{T}\}$
 - prior knowledge $\mathcal{Q}$, or $(\underline{Q}, \overline{Q})$, or contour q
- Goal: leverage prior knowledge to boost e-process's efficiency without ruining anytime-validity
- Define a *regularizer* $\rho : \mathbb{T} \rightarrow \mathbb{R}^+$ and a *regularized e-process*

$$\epsilon^{\text{reg}}(z^n, \theta) = \epsilon_\theta(z^n) \times \rho(\theta)$$

- Intuition is clear: choose $\rho(\theta)$ to be large at those θ values incompatible with prior knowledge
- e.g., $\rho(\theta)$ large if “ $\Theta = \theta$ ” is surprising

- $\rho : \mathbb{T} \rightarrow \mathbb{R}^+$ is a *regularizer* (wrt prior knowledge) if its upper expectation satisfies $\overline{Q}\rho \leq 1$
- In the possibility-theoretic case:
 - upper expectation is $\overline{Q}\rho = \int_0^1 \{\sup_{\theta: q(\theta) \geq s} \rho(\theta)\} ds$
 - so it's clear we'd want ρ big when q is small
 - basically, take " $\rho(\theta) = q(\theta)^{-1}$ "
 - that doesn't work exactly, so instead I use

$$\rho(\theta) = \{\gamma \circ q(\theta)\}^{-1}$$

where γ is a *calibrator*⁷ (Berger & Sellke, Shafer, Vovk, ...)

- If Your prior knowledge is vacuous, then take $\rho \equiv 1$

⁷Calibrators satisfy $\int_0^1 \gamma(u)^{-1} du = 1$

- Boosting the e-process jeopardizes anytime-validity... or at least the usual definition of anytime-validity
- Prior knowledge is *knowledge*, i.e., You believe it⁸
- So, You'd be willing to relax the anytime-validity target in a principled way that's consistent with Your knowledge
- My proposal:
 - lots of joint dist'ns for (Z^∞, Ω) compatible with the model and prior knowledge — any one is plausible
 - define the *upper joint dist'n* of (Z^∞, Ω) as $\bar{\mathbf{P}}$
 - replace “ $\sup_\omega P_\omega$ ” in usual def'n with $\bar{\mathbf{P}}$
- Anytime-validity wrt $\bar{\mathbf{P}}$ is *weaker* than the usual def'n

⁸You're not obligated to introduce anything You don't believe!

Generalized Ville's inequality.

$$\overline{\mathbf{P}}[\mathfrak{e}^{\text{reg}}\{Z^N, f(\Omega)\} \geq \alpha^{-1}] \leq \alpha, \quad \alpha \in [0, 1], \text{ any } N$$

Corollary.

If the regularized e-confidence set is

$$C_{\alpha}^{\text{reg}}(z^n) = \{\theta \in \mathbb{T} : \mathfrak{e}^{\text{reg}}(z^n, \theta) < \alpha^{-1}\},$$

then

$$\overline{\mathbf{P}}\{C_{\alpha}^{\text{reg}}(Z^N) \not\supseteq f(\Omega)\} \leq \alpha, \quad \alpha \in [0, 1], \text{ any } N$$

- There's an obvious *subjective* element to this proposal:
 - subjectivity isn't a dirty word!
 - necessary to maximize our contribution to science
- Proposal uses knowledge-based regularization in two ways:
 - directly boosts e-process at θ values incompatible with $\mathcal{Q}$
 - relaxes anytime-validity target so that larger ϵ 's can hit it
- Important: You don't need to compute $\overline{\mathbf{P}}$!
- Prior knowledge isn't required:
 - take $\rho \equiv 1$ when Your prior knowledge is vacuous
 - the usual anytime-validity drops out as a special case
 - but then You get no efficiency gains

- Setup

- $P_\omega = N(\omega, 1)$, write $\theta = \omega$
- original e-process is a Bayes factor

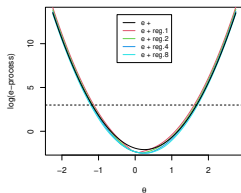
$$\begin{aligned} \epsilon_\theta(z^n) &= \frac{\int p_t(z^n) N(t \mid 0, \nu) dt}{p_\theta(z^n)} \\ &= (n\nu + 1)^{1/2} \exp\left\{-\frac{n}{2}(\theta - \bar{z}_n)^2 + \frac{1}{2}\left(\frac{n}{n\nu+1}\right) \bar{z}_n^2\right\} \end{aligned}$$

- Illustrate three different regularizations based on three different forms of prior knowledge
- *I'm not advocating for these choices,*⁹ just illustrating
- Input K varies in $\{0.1, 0.2, 0.4, 0.8\}$

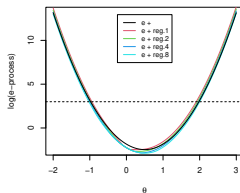
⁹I can't tell You what Your prior knowledge is!

Illustration, cont.

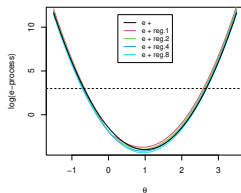
- “You expect that $|\Theta| \leq K$ ”
- $q(\theta) = 1 \wedge K|\theta|^{-1}$



(a) $\bar{z} = 0.25$



(b) $\bar{z} = 0.5$

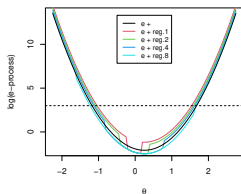


(c) $\bar{z} = 1$

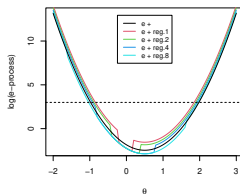
Figure: Plot of $\theta \mapsto \epsilon^{\text{reg}}(z^n, \theta)$ for three different data sets z^n .

Illustration, cont.

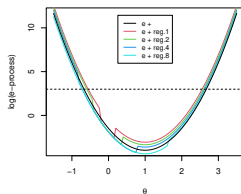
- “You’re at most $100K/5\%$ sure that $|\Theta| > 2K$ ”
- $q(\theta) = \frac{K}{5} 1(|\theta| > 2K) + 1(|\theta| \leq 2K)$



(a) $\bar{z} = 0.25$



(b) $\bar{z} = 0.5$

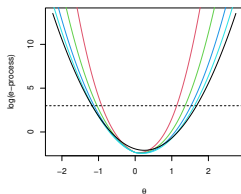


(c) $\bar{z} = 1$

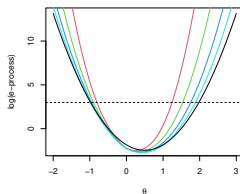
Figure: Plot of $\theta \mapsto \epsilon^{\text{reg}}(z^n, \theta)$ for three different data sets z^n .

Illustration, cont.

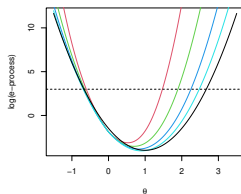
- “For each $c \in [0, 1]$, You’re $100(1 - c)\%$ sure that Θ is in the central $1 - c$ interval corresponding to a $N(0, K)$ dist’n”
- $q(\theta) = 1 - \text{pchisq}(\theta^2/K)$



(a) $\bar{z} = 0.25$



(b) $\bar{z} = 0.5$



(c) $\bar{z} = 1$

Figure: Plot of $\theta \mapsto \epsilon^{\text{reg}}(z^n, \theta)$ for three different data sets z^n .

- Regularization typically makes e-process larger, more efficient
- Exceptions are due to the calibrator: $\gamma(u) > 1$ if $u \approx 1$
- Stronger prior knowledge $\rightsquigarrow$ more efficiency gain
- Choices need to be made, in particular,
 - elicitation of prior knowledge
 - mathematizing that prior knowledge, defining q
- We prefer automation over having to make choices
 - but there's no job security in automating our contributions
 - these choices force collaboration
 - both science & statistics are stronger with collaboration

- Comparing two treatments when one may be far superior to the other can be a challenge:
 - with enough data, definitive answer can be given, but
 - unethical (?) to randomize after answer is “pretty clear”
- Hence the need for anytime-valid and efficient tests
- Now-classical example from Ware (*Stat Sci* 1989):
 - studying persistent pulmonary hypertension in newborns
 - existing treatment (CMT) not particularly effective
 - new treatment (ECMO) was very promising
- Previous studies, etc offer prior knowledge
- Opportunity for efficiency gains, ethical obligation

- Pair of unknown survival probabilities $\Theta = (\Theta_{\text{cmt}}, \Theta_{\text{ecmo}})$
- Data consists of two Bernoulli trial sequences
- Study protocol wasn't followed exactly, *stopped early*
- E-process is a type of Bayes factor¹⁰
- My encoding of Ware's prior knowledge
 - "90% confident that Θ_{ecmo} is ≥ 0.8 "
 - "50% confident that Θ_{cmt} is ≤ 0.3 "
- Mathematical encoding of marginal possibility contours:
 - $q_{\text{ecmo}}(\theta_{\text{ecmo}}) = 0.1 + 0.9 \cdot 1(\theta_{\text{ecmo}} \geq 0.8)$
 - $q_{\text{cmt}}(\theta_{\text{cmt}}) = 0.5 + 0.5 \cdot 1(\theta_{\text{cmt}} \leq 0.3)$
- Independence: $q(\theta) = q_{\text{cmt}}(\theta_{\text{cmt}}) \times q_{\text{ecmo}}(\theta_{\text{ecmo}})$

¹⁰Based on Turner & Grünwald (*SPL* 2023)

Application, cont.

- Observed data:
 - CMT 10 patients, 4 deaths
 - ECMO 9 patients, 0 deaths
- Plots of regularized and unregularized e-process contours

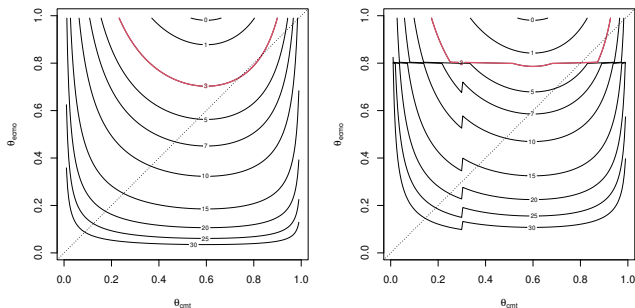


Figure: Unregularized (left) and regularized (right) e-processes

Uncertainty quantification (UQ)

- Regularized e-processes offer much more than just anytime-valid hypothesis tests & confidence sets
- Uncertainty quantification!
- Bayes/fiducial offers probabilistic uncertainty quantification about Θ — typically only asymptotically valid
- Regularized e-processes lead to *possibilistic UQ* with the following desirable properties:
 - likelihood principle is met
 - Bayesian-like no-sure-loss/coherence
 - frequentist-like anytime-validity
 - *reliability in decision-making*

- Cut to the chase...
- Define a data-driven *possibility contour*¹¹

$$\pi_{z^n}^{\text{reg}}(\theta) = 1 \wedge \mathfrak{e}^{\text{reg}}(z^n, \theta)^{-1}, \quad \theta \in \mathbb{T}$$

- Treat this as the building block of a full-blown possibilistic imprecise probability dist'n for Θ , given z^n
- Choquet integration to define upper prob/expectation:
 - $\overline{\pi}_{z^n}^{\text{reg}}(H) = \sup_{\theta \in H} \pi_{z^n}^{\text{reg}}(\theta)$
 - $\overline{\pi}_{z^n}^{\text{reg}} f = \int_0^1 \{\sup_{\theta: \pi_{z^n}^{\text{reg}}(\theta) \geq s} f(\theta)\} ds$

¹¹Grünwald (*Phil Trans Roy Soc* 2023) defines something similar

- Can mimic Bayes/fiducial with the possibilistic $\overline{\Pi}_{z^n}^{\text{reg}}$
- For example:
 - doubt truthfulness of hypothesis “ $\Theta \in H$ ” if $\overline{\Pi}_{z^n}^{\text{reg}}(H)$ is small
 - credible sets $\{\theta : \pi_{z^n}^{\text{reg}}(\theta) > \alpha\}$

Theorem.

- A test that rejects “ $f(\Omega) \in H$ ” iff $\overline{\Pi}_{z^n}^{\text{reg}}(H) \leq \alpha$ is $\overline{\mathbf{P}}$ -anytime valid
- An even stronger *uniform* Type I error bound holds,

$$\overline{\mathbf{P}}\{\overline{\Pi}_{z^n}^{\text{reg}}(H) \leq \alpha \text{ for some } H \text{ with } f(\Omega) \in H\} \leq \alpha$$

- Credible sets are $\overline{\mathbf{P}}$ -anytime-valid confidence sets

- Formal, von Neumann–Morganstern-style decision making...
- Loss function $\ell_a(\theta)$ indexed by action $a \in \mathbb{A}$
- Upper risk $a \mapsto \bar{\Pi}_{z^n}^{\text{reg}} \ell_a$, a Choquet integral
- Upper risk minimizer: $\hat{a}^{\text{reg}}(z^n) = \arg \min_a \bar{\Pi}_{z^n}^{\text{reg}} \ell_a$

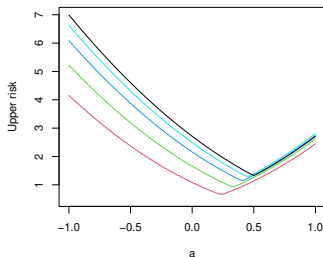


Figure: Upper risk for different priors, squared-error loss, and $\bar{z}_n = 0.5$

- Does UQ's anytime-validity offer decision-making guarantees?
- Yes!¹² In words:
 - **oracle assessment**, $\ell_a(\Theta)$, of action a is best/right
 - bad if **data-driven upper risk** is small compared to oracle's
 - theorem says the above “bad” event is $\bar{\mathbf{P}}$ -rare
- Covers the case where $\rho \equiv 1$ and $\bar{\mathbf{P}} = \sup_{\omega} P_{\omega}$

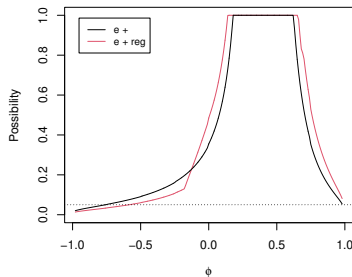
Theorem.

Among other things, the upper risk satisfies

$$\bar{\mathbf{P}}\left\{\ell_{\hat{a}^{\text{reg}}(Z^N)}(\Theta) \geq \alpha^{-1} \bar{\Pi}_{Z^N}^{\text{reg}} \ell_{\hat{a}^{\text{reg}}(Z^N)}\right\} \leq \alpha, \quad \text{all } (\alpha, N)$$

¹²See, also, M. arXiv:2112.13247

- Ware's study, revisited
- Marginal possibility contour for $\Phi = \Theta_{\text{ecmo}} - \Theta_{\text{cmt}}$
- $\Phi > 0$ or $\Phi \leq 0$?
- Lower and upper posterior expectation of Φ :
unreg $[-0.042, 0.739] \ni 0$
reg $[0.086, 0.820] \not\ni 0$

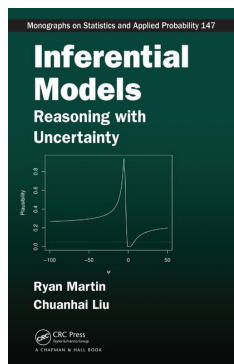


- An e-process can only be so efficient
- We're obligated to push efficiency as far as possible
- Regularization to boost efficiency?
- Two-pronged, knowledge-based regularization:¹³
 - regularizer that tends to inflate e-process
 - weakened notion of anytime-validity, allows larger e-processes
- Test/confidence procedures + full-blown possibilistic UQ
- It's taboo but, yes, I'm leaning into subjectivity...

¹³Again, I don't require users to have prior knowledge — ignorance is a special type of knowledge!

- Subjectivity is uncomfortable territory for most of us
- But this is how we maximize our contributions to science
- Relevant open questions:
 - elicitation of prior knowledge?
 - mathematical representation of knowledge?
 - most efficient calibrator?
 - is some knowledge too weak to include?
- Further research directions:
 - regularization in high-dim cases
 - prior knowledge about N

- My first book on UQ →
- New n -part series of working papers:
*Valid and efficient imprecise-probabilistic
inference with partial priors*
- $n = 3$ parts so far
 - 1 arXiv:2203.06703
 - 2 arXiv:2211.14567
 - 3 arXiv:2309.13454
- Working towards a new book on *reliable, possibilistic UQ*, related to this talk





ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

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- Papers, talks, etc — www4.stat.ncsu.edu/~rmartin/
- Questions? — rgmarti3@ncsu.edu

Thanks for your attention!